Experiment 8.1

Simulation of EMI Sources and Interference Analysis using MATLAB

Aim

To generate a desired communication signal together with different electromagnetic interference (EMI) sources in MATLAB, analyse the interference in the time and frequency domains, and measure the improvement in SNR after filtering.

Software Required

MATLAB (MATLAB Online)

Procedure / Algorithm

- Generate a 1 kHz desired signal (A = 1) sampled at fs = 10 kHz.
- Generate three EMI sources: sinusoidal EMI (0.5 V, 2.5 kHz), broadband noise and impulse noise.
- Add the EMI to the desired signal to form the received signal.
- Compute the FFT of the received signal and plot the frequency spectrum.
- Filter the received signal and calculate the SNR before and after filtering.

Output

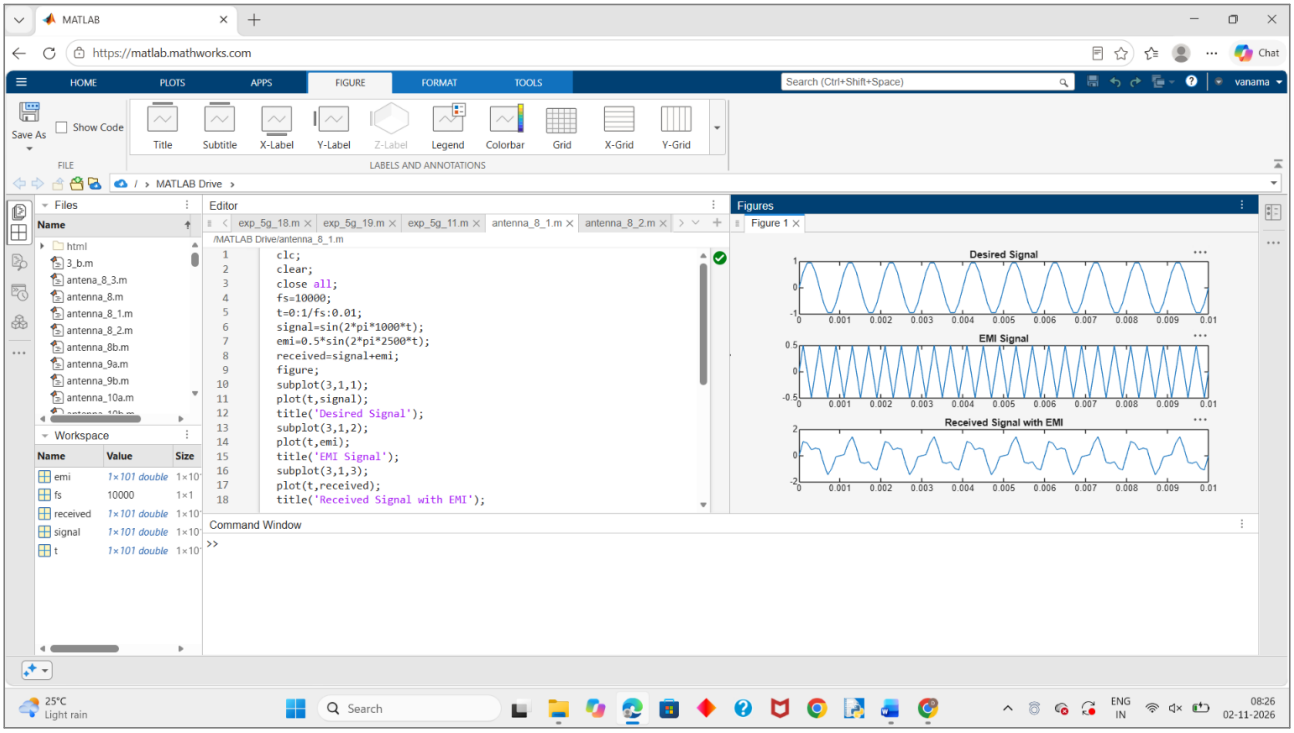


Fig. 8.1(a) Desired signal, EMI signal and received signal with EMI

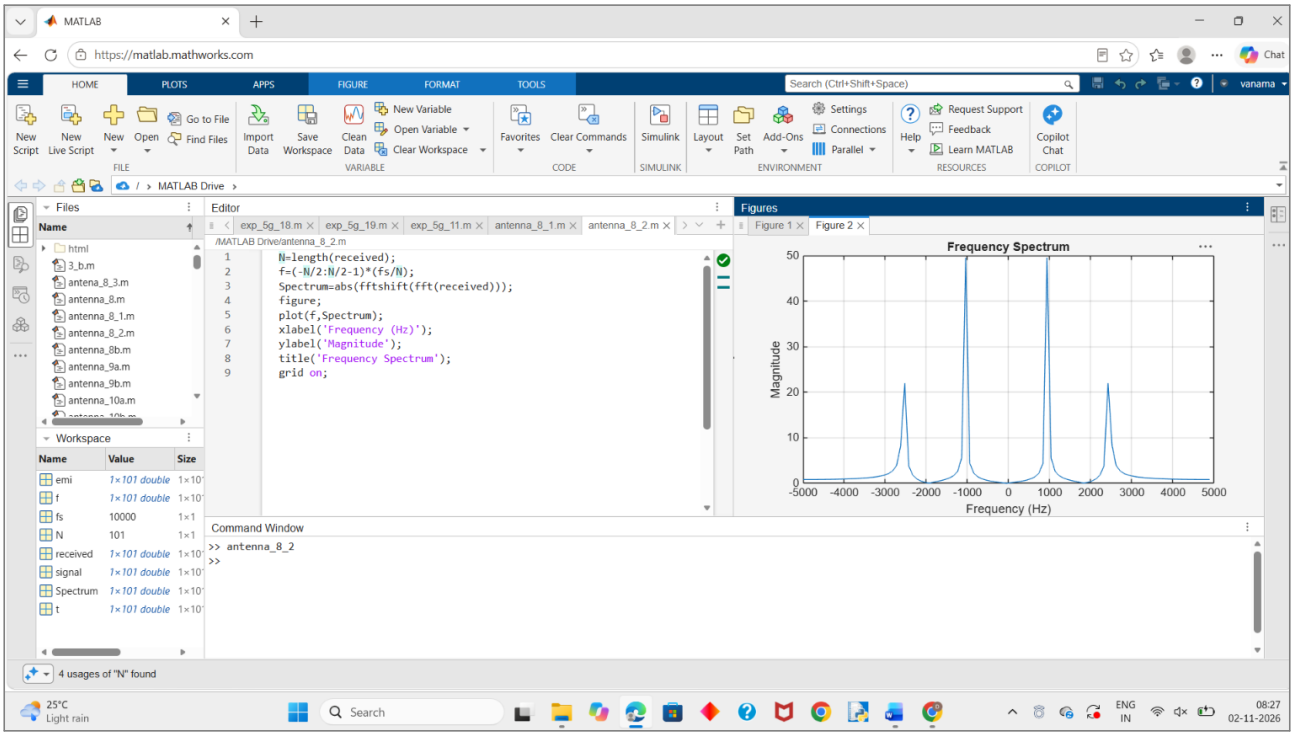


Fig. 8.1(b) Frequency spectrum of the received signal

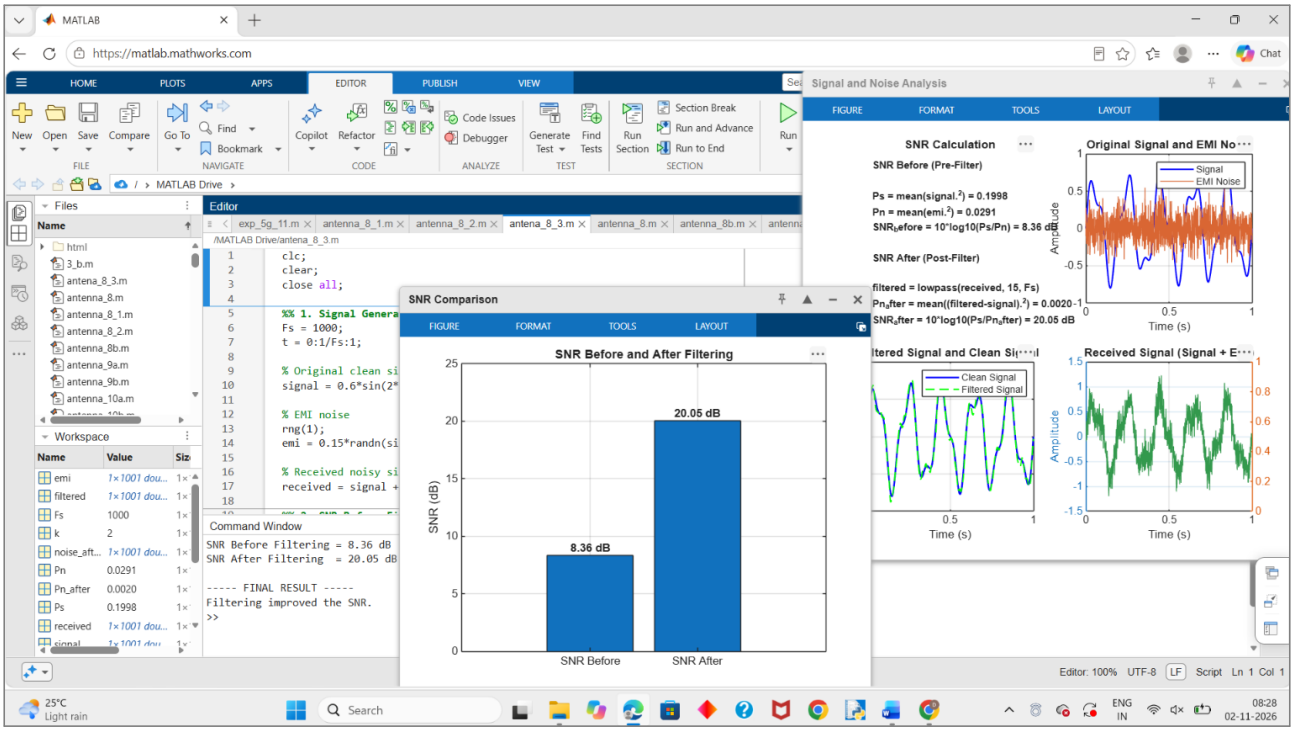


Fig. 8.1(c) SNR calculation and comparison before and after filtering

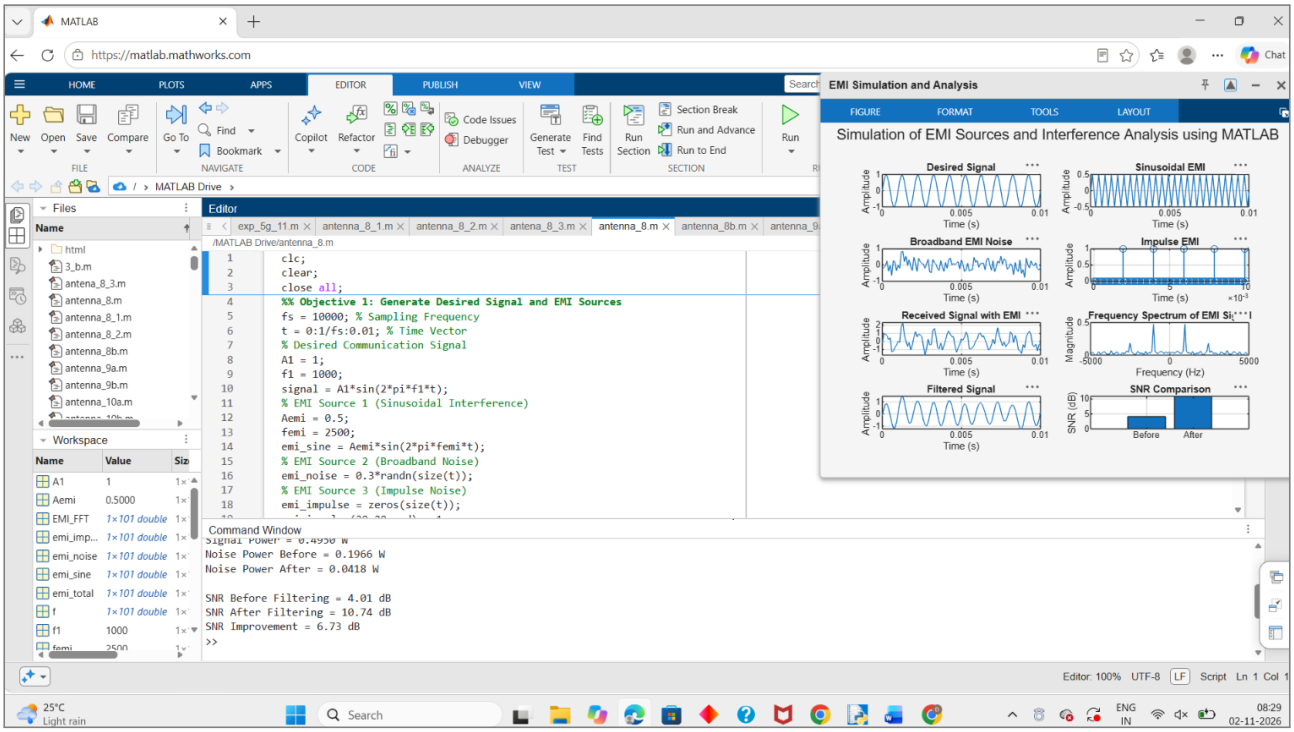


Fig. 8.1(d) Complete EMI simulation and analysis with SNR values

Result

- The frequency spectrum shows peaks at ±1000 Hz (desired signal) and ±2500 Hz (EMI), as expected.
- In the simulation with multiple EMI sources, SNR improved from 4.01 dB to 10.74 dB, an improvement of 6.73 dB.
- In the sinusoidal-plus-noise case, SNR improved from 8.36 dB to 20.05 dB after filtering.
- Thus EMI sources were simulated and the interference was analysed successfully.